

NISHANTH VEDURUVADA

Principal AI & Security Architect

Agentic AI • Enterprise AI Modernization • AI Governance • Zero Trust Architecture

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PROFESSIONAL SUMMARY

Enterprise AI and Security Architect with 9+ years across solution architecture, cloud architecture and application security. Currently AI Engineer Advisor at NTT DATA, leading architecture and technical governance for 8–10 enterprise AI modernization programs in the Syntphony product ecosystem, on platforms serving approximately 2 million users, with a team of eight engineers and data scientists. Previously sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem — work registered as an industrial design with the Government of India. Depth spans Agentic AI and multi-agent systems, Model Context Protocol (MCP) and Agent-to-Agent (A2A) tool protocols, Retrieval-Augmented Generation (RAG), Large Language Model (LLM) platforms, AI Governance, and NIST 800-207 Zero Trust Architecture, applied in regulated insurance, telecommunications and energy.

CORE COMPETENCIES

Enterprise Architecture • Solution Architecture • Cloud Architecture • Agentic AI • Generative AI • Retrieval-Augmented Generation (RAG) • AI Governance • Zero Trust Architecture • Cybersecurity • Technical Leadership

PROFESSIONAL EXPERIENCE

NTT DATA

Dec 2025 – Present

AI Engineer Advisor

Architecture and technical governance for AI modernization across NTT DATA's Syntphony ecosystem — Syntphony AI, Open Telecom Networks, Insurance Cloud Migration, Stations, and Commerce & Payments.

- **Agentic Platform Architecture:** Defined the reference architecture for a reusable multi-agent platform, standardising on MCP for tool integration and A2A for cross-domain agent collaboration, so domain teams inherit governance rather than rebuilding it per engagement.
- **AI Governance Controls:** Established per-agent OAuth2 identity, full tool-call audit logging, configurable rate limits, action-scope enforcement, and a human-in-the-loop approval gateway for high-risk actions — inherited automatically by every domain agent on the platform.
- **Enterprise Retrieval Architecture:** Designed a dual-path knowledge assistant separating live analytics from indexed content: semantic-model queries with row-level security passed through as the calling user, alongside permission-scoped document retrieval, routed by a query classifier. Metric values are never embedded, so answers cannot go stale.
- **Legacy Modernization:** Architected a RAG system over mainframe COBOL and JCL estates for insurance cloud migration, built on a structure-aware parser and static dependency graph. Chose retrieval over fine-tuning so every answer cites program, paragraph and line.
- **Cross-Domain Operational Intelligence:** Designed multi-agent orchestration across five OSS domains — RAN, Core, Transport, Inventory and Assurance — using domain-specialist agents under a query planner, with a correlation layer producing evidence-cited answers rather than model summaries.
- **Conversational Analytics:** Designed an embedded analytics assistant over existing semantic models, with multi-tenant isolation via Entra ID and row-level security scoped per business unit and country, avoiding duplication of the client's existing BI investment.
- **Technical Leadership:** Lead architecture reviews, decision records and trade-off analysis for a team of eight; own enterprise AI architecture standards, technical strategy, stakeholder alignment and mentoring.

Levo.ai

Jan 2025 – Dec 2025

AI Enterprise Architect

Client: Insurance Information Bureau of India (IIB)

Sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem aggregating records from every insurance company in India, under IRDAI regulation and data-localisation requirements.

- **Zero Trust Architecture:** Designed the full NIST 800-207 structure — policy administration, decision, enforcement and information points as separate concerns — with enforcement deployed as an API gateway intercept so every call, human or agent, traverses it.
- **Adaptive Trust Engine:** Replaced binary allow/deny with continuous trust scoring across identity confidence, device posture, behavioural baseline, data classification and context, driving allow / step-up-authentication / deny decisions with thresholds configurable per data class.
- **AI Agent Identity & Governance:** Gave agents their own workload identity rather than borrowed user credentials, with action-scope enforcement, per-agent rate limiting, prompt-injection detection, full decision audit, and human escalation above a risk threshold.
- **Offline AI Capability:** Architected fully offline inference with Ollama to satisfy no-internet-egress environments — a hard constraint that would otherwise have blocked the entire governance layer.
- **RAG Policy Intelligence:** Built natural-language access to IRDAI regulations and internal policy, orchestrated with LangGraph and CrewAI, with grounded and cited responses for compliance analysts.

- **Explainability:** Wired traceable rationale into every access decision — which policy matched, which trust signals fired — making AI-assisted decisions defensible under regulatory audit.

Arcstream Technologies

Aug 2022 – Jan 2025

Technical Security Lead

- **Enterprise Security Modernization:** Led Zero Trust network segmentation and a SASE migration covering 5,000+ users, replacing perimeter-based access with identity-centric controls.
- **Secure Delivery:** Hardened CI/CD pipelines and established architecture governance across engineering teams.

Reezanorp Technologies

Dec 2016 – Jul 2022

Senior Software Engineer

- **Applied ML & Data Engineering:** Built machine learning, analytics, computer vision, ETL and enterprise automation systems in Python — the engineering foundation underneath the later architecture work.

PUBLICATIONS & INTELLECTUAL PROPERTY

MCP Guardian: A Security-First Layer for Safeguarding MCP-Based AI Systems

Computer Science & Information Technology (CS&IT), Vol.15 No.9, pp.107–121, 2025 · DOI 10.5121/csit.2025.150908

Middleware for the Model Context Protocol adding authentication, per-token rate limiting, WAF-style scanning of tool calls and request logging, without modifying individual MCP servers. Evaluated at roughly 3–4 ms median latency overhead.

AI Trust Fabric — Secure Insurance Data Management Device

Registered Design No. 485611-001, Government of India · Co-inventor

The Zero Trust and AI governance architecture designed for a national-scale insurance data ecosystem — NIST 800-207 policy separation, adaptive trust evaluation, identity-centric controls and secure data access mechanisms.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, Multi-Agent Systems, Generative AI (GenAI), Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Machine Learning, Predictive Analytics, Computer Vision, Conversational AI, Semantic Search, Prompt Engineering, LLMops, AI Evaluation

AI Frameworks & Protocols: Model Context Protocol (MCP), FastMCP, Agent-to-Agent (A2A), CrewAI, LangGraph, LangChain, AutoGen, Semantic Kernel, Tool Calling, Human-in-the-Loop, Vector Databases, ChromaDB, Azure AI Search

AI Platforms: Azure OpenAI, Azure AI Foundry, OpenAI, Anthropic Claude, Ollama, Google Vertex AI, GPT-4o

Security & Governance: Zero Trust Architecture, NIST 800-207, Policy Administration/Decision/Enforcement/Information Points (PAP/PDP/PEP/PIP), Identity and Access Management (IAM), Microsoft Entra ID, OAuth 2.0, OpenID Connect (OIDC), Multi-Factor Authentication (MFA), Zero Trust Network Access (ZTNA), Secure Access Service Edge (SASE), Data Loss Prevention (DLP), API Security, AI Governance, Explainability, Audit & Compliance, Row-Level Security (RLS)

Cloud & Platform: Microsoft Azure, Kubernetes (AKS), Docker, Microservices, Event-Driven Architecture, Apache Kafka, Azure Service Bus, Azure API Management, Azure Logic Apps, Cosmos DB, Microsoft Fabric, Power BI, Microsoft Graph, OpenTelemetry, CI/CD, DevSecOps

Engineering: Python, FastAPI, REST APIs, GraphQL, SQL, Git, Legacy Modernization, COBOL/JCL analysis, ETL, Solution Architecture, Enterprise Architecture, Technical Leadership, Mentoring, Stakeholder Management

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering — Jawaharlal Nehru Technological University (JNTU), India